$$\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{vmatrix} u_{31} & u_{32} & u_{33} \end{vmatrix} \begin{pmatrix} x_3 \end{pmatrix} = \begin{pmatrix} u_{13}x_1 + u_{23}x_2 + u_{33}x_3 \end{pmatrix} \begin{pmatrix} x_3 \end{pmatrix}$$

$$\begin{vmatrix} u_{11}x_1 + u_{21}x_2 + u_{31}x_3 & u_{12}x_1 + u_{22}x_2 + u_{32}x_3 & u_{13}x_1 + u_{23}x_2 + u_{33}x_3 \end{vmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} =$$

$$u_{11}x_1 + u_{21}x_1x_2 + u_{31}x_1x_3 + u_{12}x_1x_2 + u_{22}x_2 + u_{32}x_2x_3 + u_{13}x_1x_3 + u_{23}x_2x_3 + u_{33}x_3$$

$$u_{11} + u_{21} + u_{31} + u_{12} + u_{22} + u_{32} + u_{13} + u_{23} + u_{33} \Rightarrow \text{quadratic term}$$

$$u_{11} \rightarrow \text{Coefficient of } x_1^2$$
$$u_{22} \rightarrow \text{Coefficient of } x_2^2$$
$$u_{33} \rightarrow \text{Coefficient of } x_3^2$$

$$u_{12} + u_{21} \Rightarrow u_{12} = u_{21} = \frac{1}{2}\text{coefficient of } x_1x_2$$
$$u_{13} + u_{31} \Rightarrow u_{13} = u_{31} = \frac{1}{2}\text{coefficient of } x_1x_3$$
$$u_{23} + u_{32} \Rightarrow u_{23} = u_{32} = \frac{1}{2}\text{coefficient of } x_2x_3$$

Three Variables Example

$$\begin{array}{c|ccc} & x_1 & x_2 & x_3 \\ \hline x_1 & u_{11} & u_{12} & \\ x_2 & u_{21} & u_{22} & \\ x_3 & u_{31} & u_{32} & \end{array}$$

$$\begin{array}{|c|c|c|} \hline \text{coef of } x_1^2 & \frac{1}{2}\text{coef of } x_1x_3 & \frac{1}{2}\text{coef of } x_1x_2 \\ \hline \frac{1}{2}\text{coef of } x_1x_2 & \text{coef of } x_2^2 & \frac{1}{2}\text{coef of } x_2x_3 \\ \hline \frac{1}{2}\text{coef of } x_1x_3 & \frac{1}{2}\text{coef of } x_2x_3 & \text{coef of } x_3^2 \\ \hline \end{array}$$